

Multi-Dialect Marathi Text Normalization via LLM-Driven Synthetic Augmentation

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Abstract

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Introduction

Natural Language Processing (NLP) has achieved significant advances in recent years, allowing computers to analyze, interpret, and generate human language. Modern architectures for both, understanding and generation of natural language are based on models built upon Transformers architecture ???. Large Language Models (LLMs) built upon this architecture had been adopted for various NLP tasks across diverse real-world problems in healthcare, finance, and legal infrastructure ???.

NLP has faced several problems in incorporating low-resource languages, especially regional dialects of higher-resource languages ???. Surveys among native dialect communities reveal that proper analysis and understanding of dialectal text remains a priority ?. highlighting a need for converting non-standard dialect into a standard form while keeping the original meaning intact ?. Advancing NLP methods have made it possible to design systems to seamlessly link previously inaccessible low-resource regional dialects.

India has constitutionally recognized 22 languages with hundreds of regional dialects spoken by over 1.4 billion people. While languages such as Hindi, Bengali, Marathi, Tamil, Telugu, etc. have started receiving attention through initiatives like AI4Bharat IndicCorp and BPCC ?, sub-regional dialects remain almost entirely untouched in NLP. One of the examples of this linguistic ambiguity is Marathi, an Indo-Aryan language spoken by over 83 million people, primarily in Maharashtra ?. Marathi is written in Devanagari from left to right and stands as one of India's 22 constitutionally recognized scheduled languages. While Standard Marathi (based on the dialect variety of Pune city and surrounding regions) has a well-defined literary standard and established digital infrastructure, regional sub-dialects tend to show phonological, and lexical divergence from the standard form.

When users speak or write in the regional dialects, existing NLP systems such as Automatic Speech Recognition (ASR), Machine Translation (MT), conversational AI agents, and information retrieval tools fail severely due to high Out-Of-Vocabulary (OOV) rates, non-standard verbal forms, and phonetic spelling variations. *Dialect Normalization* is the task of automatically converting non-standard dialectal text into normative standard written form while strictly preserving underlying semantic intent therefore becomes a critical prerequisite for digital inclusion.

Developing systems for dialect normalization have two fundamental challenges: (1) *Extreme Data Scarcity*: datasets connecting regional Indic dialects to standard written forms are

non-existent; and (2) *Morphological Complexity*: Marathi is a morphologically rich language where case markers and verbal tense/aspect suffixes require fine-grained subword transformation rather than a simple dictionary lookup.

To address these challenges, this paper presents a multi-dialect Marathi text normalization framework combining automated synthetic parallel data augmentation with sequence-to-sequence transformer fine-tuning. Our main contributions are :

- **Parallel Marathi Dialect Corpus with Synthetic Augmentation**: Based on the RESPIN corpus [1], we curate and expand a parallel corpus of verified sentence pairs across Malvani, Ahirani, and Varhadi dialects mapped to Standard Marathi across domains of agriculture and finance, combining human-translated pairs with an automated synthetic generation and multi-level verification pipeline.
- **End-to-end Fine-Tuning Pipeline**: We utilize a neural fine-tuning framework for dialect normalization, adapting pre-trained multilingual sequence-to-sequence transformer models for standardization.
- **Empirical Evaluations**: We implement a standardized train-test split combined with cross-validation across distinct dataset configurations, conducting experiments, verification filtering analyses, and error evaluations.

Related Work

Dialect Normalization and Text Standardization

Dialect Normalization has been established as a sentence-level character transduction task alongside historical and user-generated content normalization [2].

Early neural explorations by Partanen et al. [3] evaluated sequence transduction across 23 Finnish dialect varieties, demonstrating that sliding-window chunk-level neural architectures (LSTMs and character Transformers) effectively normalize non-standard dialectal forms into normative Standard Finnish, drastically reducing word error rates from an initial 52.89% down to 5.73%. Kuparinen et al. [4] extended this paradigm to a large-scale multilingual benchmark covering four distinct language families (Finnish, Norwegian, Swiss German, and Slovene), showing that fine-tuned sequence-to-sequence transformers and pre-trained byte-level architectures (byT5) consistently surpass character-level statistical machine translation (SMT) and recurrent baselines across diverse dialectal corpora.

In the Arabic NLP community, the Conventional Orthography for Dialectal Arabic (CODA) framework [5] established foundational orthographic guidelines for mapping diverse regional Arabic varieties (Egyptian, Levantine, Gulf, Maghrebi) to Modern Standard Arabic (MSA), providing the linguistic substrate for parallel corpus creation and computational Arabic dialect processing. The MADAR project subsequently extended these principles to fine-grained city-level dialect identification and translation across 25 Arab cities. While early computational approaches relied on character-level finite-state transducers (FSTs) and phrase-based SMT rules, contemporary work predominantly leverages pre-trained multilingual encoder-decoder transformers.

In Germanic languages, Swiss German dialect normalization to Standard High German has been investigated extensively due to the total absence of a single standardized orthography for Swiss German dialects. Early studies explored character-level statistical rules, while recent work by Kresic and Abbas [6] demonstrated that fine-tuning multilingual sequence-to-sequence transformers on the SwissDial corpus yields high-quality text normalization, with the compact mT5-small model achieving competitive performance against significantly larger base and large transformer variants.

For Indic languages, dialect normalization remains a critically underexplored area. While prior literature has addressed code-mixing (e.g., Hindi-English) and script transliteration, mapping regional dialectal text to a standardized written form within the same language has received

Table 1: Comparison with Prior Marathi Dialect Processing Research

[c]Study Size (%) Data	[c]Task [c]Best Acc. [c]Open	[c]Dialects	[c]Data			
Mulla & Kumar (2022) ?	Acoustic Classification	4	7,555	84.24	No	
Pawar et al. (2026) ?	Acoustic Identification	3	—	93.65	No	
Gaikwad et al. (2025) ?	Zero-Shot Translation	1	RAG	—	No	
Vyavhare et al. (2026) ?	Hybrid Rule+Neural	—	—	—	No	

minimal computational attention. A comprehensive global survey by Joshi et al. ? categorizes dialectal NLP tasks across Natural Language Understanding (NLU) and Natural Language Generation (NLG) across diverse language families, emphasizing the acute performance degradation of state-of-the-art models on non-standard dialects and calling for equitable language technologies. Recently, Dimakis et al. ? introduced a hybrid normalization framework combining rule-based morphological transformations with LLM few-shot prompting to standardize low-resource Greek dialect proverbs without requiring prior parallel training data.

In the specific context of Marathi language dialect processing, research has historically concentrated on speech, acoustic classification, or localized applications:

- **Acoustic Speech Classification & Recognition:** Mulla and Kumar ? explored spectro-temporal feature extraction (15 parameters including chromagram, spectral centroid, and MFCCs) with Chi-Square/ANOVA feature selection on 7,555 speech recordings from the LDC-IL Marathi raw speech corpus across four dialects (Goa-based, Vidarbha, Marathwada, Puneri), achieving 84.24% accuracy with Ridge Classifiers. Pawar et al. ? investigated acoustic dialect identification across Ahirani, Varhadi, and Standard Marathi audio using MFCC features and Convolutional Neural Networks (CNNs) under clean and artificially generated Gaussian noise conditions (SNR 1.78–9.74 dB), achieving up to 93.65% accuracy in clean environments and 87.29% under noisy conditions. Amartyaveer et al. ? and Kumar et al. ? at IISc Bangalore introduced RESPIN-S1.0 (a 10,000+ hour read-speech corpus across 38 Indian dialects and 9 languages, including 4 Marathi sub-dialects: Southern Konkani, Northern Konkani, Standard Marathi, and Varhadi) and developed multimodal ASR and RoBERTa embeddings for dialect identification, achieving 79.81% accuracy.
- **Text-Level Dialect Processing & Translation:** Gaikwad et al. ? explored zero-shot prompting via the commercial Gemini API within a Retrieval-Augmented Generation (RAG) framework using Sentence Transformers and FAISS to translate Konkani (Devanagari) dialect queries into Standard Marathi for document Q&A. Vyavhare et al. ? proposed a hybrid rule-based dictionary and neural translation web interface for regional Marathi dialects. Gavali ? introduced a student-level framework combining IndicBERT and IndicWav2Vec for dialect detection and normalization on small local datasets.

Our work significantly advances this landscape: while prior studies focused either on speech/acoustic classification or API-dependent zero-shot prompting, we establish the first large-scale parallel-pair text benchmark, introduce an automated LLM synthetic data expansion engine (powered by Llama-3.1-8B ? and Gemma ?) with multi-tier verification, and conduct rigorous 5-fold cross-validated fine-tuning of open-weights sequence-to-sequence transformers (mT5-small and IndicBART) for local, reproducible inference.

Table 1 summarizes the key distinctions between our work and prior approaches to Marathi dialect processing.

Indic NLP and Sequence-to-Sequence Transformers

The development of pre-trained multilingual models for Indic languages has accelerated significantly in recent years. IndicBART [1] introduced a multilingual sequence-to-sequence model based on the mBART-50 architecture [2], pre-trained on 11 Indic languages using a unified Devanagari script representation with a 64,000-token SentencePiece vocabulary. The model leverages orthographic similarity between Indic scripts to facilitate cross-lingual transfer learning and demonstrated competitive performance on neural machine translation and extreme summarization tasks despite being significantly smaller than general-purpose multilingual models.

Concurrently, general multilingual models such as mT5 [3]—the multilingual extension of T5 [4]—extended Text-to-Text Transfer Transformers across 101 languages using the mC4 (Multilingual Colossal Clean Crawled Corpus). mT5 features a 250,000-token SentencePiece vocabulary and demonstrated state-of-the-art performance on the XTREME multilingual benchmark. Its text-to-text formulation, which casts all NLP tasks as input-output text transformations, makes it particularly well-suited for sequence-level normalization tasks.

The L3Cube-MahaNLP initiative [5] has contributed Marathi-specific transformer models (MahaBERT, MahaRoBERTa) and curated datasets for tasks including sentiment analysis, named entity recognition, and hate speech detection. However, these encoder-only models are designed for classification tasks rather than sequence-to-sequence generation required for dialect normalization.

Synthetic Data Generation for Low-Resource NLP

The use of Large Language Models (LLMs) for synthetic data generation has emerged as a powerful strategy for overcoming data scarcity in low-resource settings. Recent work has demonstrated that LLMs function more effectively as data generators than as direct task solvers, enabling a “generate-then-fine-tune” paradigm where synthetic parallel data from a large teacher model is used to train smaller, task-specific student models [6]. Key considerations in this approach include maintaining semantic fidelity, preventing hallucination, and implementing multi-tier verification pipelines to ensure data quality [7].

For Indic languages specifically, synthetic data augmentation has been applied primarily to machine translation tasks, with back-translation and pivoting through high-resource languages serving as common strategies. Our work extends this paradigm specifically to the dialect normalization task, introducing domain-aware generation templates and multi-stage LLM verification protocols tailored to the morphological properties of Marathi dialects.

Methodology

Target Dialects and Linguistic Properties

We target three major Marathi dialects mapped in social dialectology and comparative linguistic surveys of Maharashtra [8] that collectively span the primary geographic regions of Maharashtra and exhibit distinct linguistic transformation patterns:

1. **Malvani (Konkan Coast):** Characterized by phonetic shortening of word-final vowels, palatalization of dental consonants, and distinct auxiliary verb forms. Lexical transformations include systematic vowel shifts: standard जातो (“he goes”) becomes Malvani जातां; standard कुठे (“where”) becomes कुठें. Domain-specific terminology in agriculture and banking is generally preserved but often accompanied by Konkani-influenced function words.
2. **Ahirani (Khandesh Region):** Heavily influenced by neighboring Gujarati and Bhili language families, exhibiting significant consonant cluster simplification and retroflex-dental consonant shifts. Characteristic transformations include: standard कशाला (“why”) → Ahirani कसालें; standard मुलगा (“boy”) → Ahirani पोर. Ahirani presents the highest degree of

Table 2: RESPIN-S1.0 Marathi Source Corpus Statistics

Statistic	Value
Total Utterances	809,934
Total Duration	1,026.28 hours
Unique Speakers	2,644
Unique Pincodes	234
Gender Distribution	51% Female, 49% Male
Domains	Agriculture, Banking
Domain Split	49% Agriculture, 51% Banking
Dialect Variants	Malvani, Ahirani, Varhadi

lexical divergence among the three target dialects.

3. **Varhadi (Vidarbha/Eastern Maharashtra):** Spoken in Vidarbha, exhibiting characteristic interrogative transformations and systematic verb suffix modifications. Standard interrogative का? (“why?”) becomes Varhadi काऊन?; verbal forms undergo regular suffix mapping patterns. Among the three dialects, Varhadi maintains the highest degree of structural similarity to Standard Marathi, with transformations being more systematic and rule-governed.

The target dialects display rich phonological, morphological, and lexical transformation paradigms across representative sentence structures alongside Standard Marathi.

Original Corpus Curation and Foundation in RESPIN

The foundational dialectal sentence prompts for our dataset were drawn from the domain-specific text transcripts of the RESPIN-S1.0 initiative [?], developed at IISc Bangalore under the Gates Foundation project. While RESPIN provides audio recordings and mono-lingual text prompts across agriculture and banking domains for Indian regional dialects (including Marathi sub-dialects: Southern Konkan/Sindhudurg, Northern Konkan/Ahirani, and Varhadi/Nagpur), it lacks aligned parallel Standard Marathi target translations required for sequence-to-sequence text normalization.

To construct a supervised normalization benchmark, native Marathi linguists performed sentence-level parallel translation of RESPIN dialectal prompts into normative Standard Marathi (मानक मराठी), assembling an initial clean parallel corpus of 16,163 pairs across agriculture and banking domains. The initial corpus comprises Malvani (5,576 pairs), Ahirani (5,501 pairs), and Varhadi (5,086 pairs). A comprehensive manual audit protocol flagged critical translation flaws including garbled syntax (incomplete or malformed sentences), semantic inversions (meaning reversal in normalization), Hindi language leakage (Standard Hindi substituted for Standard Marathi), retained dialect residue (incompletely normalized outputs), and dropped question marks (loss of interrogative intent). All flagged pairs were manually corrected and validated before seeding the synthetic expansion pipeline.

RESPIN Source Corpus Statistics

The RESPIN-S1.0 Marathi partition, from which our dialect sentence prompts were sourced, represents a large-scale collection of read speech across rural Maharashtra. Table 2 summarizes key corpus statistics.

The large speaker diversity (2,644 unique speakers across 234 pincodes) ensures broad coverage of intra-dialect phonetic and lexical variation, while the balanced gender and domain splits reduce systematic bias in our derived parallel corpus.

Table 3: Baseline ASR Performance (indic-conformer-600m) Dialectwise Breakdown on IISc_RESPIN_test_mr

[l]Dialect Name (h)	[c]Utts	[c]Duration			
WER (%)	[c]Norm.				
CER (%)	[c]Norm.				
Match (%)	[c]Exact				
Malvani	559	0.78	34.37	7.45	21.47
Ahirani	540	0.76	24.14	5.08	31.85
Varhadi	516	0.72	24.79	5.14	30.62
Standard Marathi	555	0.78	10.87	2.34	39.10

IndicConformer Baseline ASR Evaluation

To establish a comprehensive speech-and-language baseline, we evaluate pre-trained automatic speech recognition (ASR) performance on the official IISc_RESPIN_test_mr test benchmark set, comprising 2,170 speech utterances (3.04 hours of recorded audio) across all four regional sub-dialects: Southern Konkani (Malvani), Northern Konkani (Ahirani), Standard Pune Marathi, and Varhadi (Vidarbha).

We evaluate ai4bharat/indic-conformer-600m-multilingual under both Connectionist Temporal Classification (CTC) and Recurrent Neural Network Transducer (RNN-T) decoding setups. Overall on 2,170 utterances, CTC decoding achieves 23.94% WER (5.12% CER, 30.12% Exact Match), while RNN-T decoding achieves optimal baseline performance with 23.71% WER, 5.06% CER, and 30.78% Exact Match.

Table 3 details the dialectwise ASR baseline performance across all four regional sub-dialects using the primary RNN-T decoder.

Sub-dialect analysis reveals significant acoustic-phonetic difficulty variation across regional varieties:

- **Standard Marathi:** Achieves the lowest baseline error at 10.87% WER and 2.34% CER (555 utterances).
- **Ahirani:** Achieves 24.14% WER and 5.08% CER (540 utterances).
- **Varhadi:** Achieves 24.79% WER and 5.14% CER (516 utterances).
- **Malvani:** Proves to be the most challenging dialect for speech recognition, exhibiting 34.37% WER and 7.45% CER (559 utterances) due to distinct vocalic shortening and phonetic contractions.

LLM-Driven Synthetic Generation Architecture

To address data scarcity, we developed an automated synthetic parallel data generation pipeline powered by Gemma-2 ? in both 9B-instruction-tuned (gemma-2-9b-it) and 27B-instruction-tuned (gemma-2-27b-it) configurations, accessed via the Google AI Studio API. The pipeline architecture incorporates several design decisions to ensure generation quality:

- **1-Prompt-Per-Sentence Architecture:** Rather than generating batches of parallel pairs per API call (which risks array length mismatches and cross-sentence context contamination), each dialectal sentence is processed individually with a self-contained prompt.
- **Zero-Hallucination Prompting:** The generation prompt enforces strict semantic fidelity rules: the standard output must preserve exact meaning, domain terminology (agriculture/banking), named entities, and numerical values from the dialectal input. The prompt

Table 4: Parallel Dataset Composition and Yield Across All Three Dialect Suites

[c]Dialect Name Pairs Pairs Pairs	[c]Original [c]Synthetic [c]Total		
Malvani	5,576	5,569	11,145
Ahirani	5,501	5,534	11,035
Varhadi	5,086	5,069	10,155
Total (All 3 Dialects)	16,163	16,172	32,335

includes dialect-specific lexical mapping tables and 5–8 few-shot reference examples per dialect.

- **Temperature Control:** Generation temperature is set to 0.3 with top- $p = 0.9$ to balance diversity with fidelity. Multiple candidate generations are produced and ranked by verification scores.
- **Domain Preservation:** Generation prompts explicitly enumerate 100% preservation rules for domain-specific entities including crop names (शेती, कापूस, ज्वारी), banking terminology (कर्ज, खाते, व्याजदर), and geographical references.

Multi-Tier Verification and Quality Filtering

All generated pairs undergo a rigorous three-tier verification and correction pipeline:

1. **Tier 1: Rule-Based Syntactic Filter.** Automated validation checks implement Indic Abugida script normalization principles ?, including:
 - *Devanagari Script Integrity:* Both source and target must contain >90% Devanagari Unicode characters within the U+0900--U+097F range, automatically normalizing broken nuktas (U+093C), invalid halants (U+094D), and orphan matras.
 - *Length Ratio Constraint:* $0.5 \leq |S_{\text{dialect}}|/|S_{\text{standard}}| \leq 2.0$ to eliminate hallucinated sentence expansions or accidental truncations.
 - *Non-Identity Constraint:* $S_{\text{dialect}} \neq S_{\text{standard}}$ to exclude trivially un-normalized identical string copies.
 - *Minimum Token Count:* Both source and target must contain ≥ 3 whitespace-separated tokens.
2. **Tier 2: LLM Semantic Verification.** A secondary LLM evaluator using llama-3.1-8b-instant (accessed via Groq API with key rotation pool and NVIDIA NIM fallback for rate limit resilience) performs semantic equivalence verification. The verifier rates candidates on a 1–5 score scale assessing: (a) exact semantic preservation; (b) absence of hallucinated entities; (c) absence of Hindi/English contamination; and (d) complete normalization of dialectal inflections. Pairs scoring < 4.0 are rejected or routed to correction.
3. **Closed-Loop Correction Engine.** Pairs flagged by Tier 1 or Tier 2 enter a two-step correction loop: Step 1 generates corrected candidate translations using feedback-aware prompts highlighting the specific failure reason (e.g., “Hindi leakage detected” or “Incomplete normalization”); Step 2 (QA Auditor) re-verifies corrected candidates, discarding pairs that fail a second audit pass.

As detailed in Table 4, the synthetic augmentation pipeline contributed 16,172 high-quality verified parallel pairs, nearly perfectly doubling the original corpus to produce an expanded